Chapter 15

Pharmacovigilance Systems

Dixon Thomas[1] and Christoph Klika[2]
[1]Gulf Medical University, Ajman, United Arab Emirates; [2]University of Duisburg-Essen, Duisburg, Germany

Learning Objectives:

Objective 15.1 Introduce the concepts pharmacovigilance, adverse drug event, and adverse drug reaction.
Objective 15.2 Describe the pharmacovigilance framework and the roles of different stakeholders in patient safety.
Objective 15.3 Explain how to report adverse drug events.

OBJECTIVE 15.1. INTRODUCE THE CONCEPTS PHARMACOVIGILANCE, ADVERSE DRUG EVENT, AND ADVERSE DRUG REACTION

Pharmacovigilance (WHO 2002)

The World Health Organization (WHO) defines pharmacovigilance as "the science and activities relating to the detection, assessment, understanding and prevention of adverse effects or any other drug-related problem."[1]

Pharmacovigilance has four general objectives:

- to improve patient care and safety in relation to the use of drugs, and all medical and paramedical interventions;
- to improve public health and safety in relation to the use of drugs;
- to contribute to the assessment of benefit, harm, effectiveness, and risk of drugs, encouraging their safe, rational, and more effective (including cost-effective) use;
- to promote understanding, education, and clinical training in pharmacovigilance and its effective communication with health professionals and the public.

Pharmacovigilance is an essential part of the healthcare system in many countries. As part of national healthcare, pharmacovigilance is regulated through national laws on drug policy. In the European Union (EU), the national laws of member countries have to conform with drug policies made in the EU at the regional level. The pharmacovigilance definitions may vary in some countries in relation to the differences in drug laws. Most of the countries follow more or less the same definitions of the WHO. The WHO in collaboration with the Council of International Organizations of Medical Sciences (CIOMS) and the International Conference on Harmonization (ICH) compiled key terms in pharmacovigilance.[1]

Pharmacovigilance is not just a science of adverse drug reactions (ADRs). It is one of the main sciences dealing with patient safety and includes any drug-related problem that results in adverse events. Pharmacovigilance is an essential part of the drug development process and continues to be important in clinical practice through day-to-day responsibilities of healthcare professionals and postmarketing research. No drug is approved and authorized for use on the market without assessing pharmacovigilance data. Safe and effective use of drugs is fundamental in clinical policy. The question to be asked here is whether the issue of drugs' ineffectiveness should be dealt with by pharmacovigilance. The answer is yes. A drug is supposed to be effective in decreasing the burden of disease; if it fails to do so, harm to the patient is caused by treatment failure, which is a drug-related problem.

Adverse event: Medical occurrence temporally associated with the use of a medicinal product, but not necessarily causally related. Adverse events are commonly called adverse drug events (ADEs).

One of the biggest challenges in pharmacovigilance is to separate the harm caused by the disease from that caused by the drug. When the cause of the reaction could not be associated with the medicinal product, it is an ADE. An ADE after

Clinical Pharmacy Education, Practice and Research. https://doi.org/10.1016/B978-0-12-814276-9.00015-5
Copyright © 2019 Elsevier Inc. All rights reserved.

taking a drug may be a complication or altered pathogenesis of the disease itself. There is also the possibility that mysterious genetic or environmental factors are involved. In polypharmacy, the matter becomes more complicated because it is sometimes difficult to identify which drug causes the reaction.[10–19]

Adverse reaction (WHO 1972): A response to a drug that is noxious and unintended, and that occurs at doses normally used in humans for the prophylaxis, diagnosis, or therapy of disease or for the modifications of physiological function.

An adverse reaction, commonly called adverse drug reaction (ADR), has an established causal relationship of a drug or a combination of drugs to the reaction caused. The so-called causality assessment describes the systematic evaluation of reported ADRs to establish a causal link between a drug and the adverse event. The ADR is not necessarily a clinical symptom like arash; it could also be a significant variation in a laboratory test or other diagnostic measures. An ADR is something that happens when a drug is administered in normal doses to a consumer, not a reaction caused by overdosing (intentional or unintentional). Note that the definition of pharmacovigilance in the EU is different because ADRs include not only overdose but also off-label use, drug abuse, and medication errors.[2] In general, however, ADRs exclude reactions attributed to drug abuse or allergic reactions happening to a particular patient. Some definitions are more inclusive, e.g., the American Society of Health-System Pharmacists defines a significant ADR as any unexpected, unintended, undesired, or excessive response to a drug that

1. Requires discontinuing the drug (therapeutic or diagnostic),
2. Requires changing the drug therapy,
3. Requires modifying the dose (except for minor dosage adjustments),
4. Necessitates admission to a hospital,
5. Prolongs a stay in a healthcare facility,
6. Necessitates supportive treatment,
7. Significantly complicates diagnosis,
8. Negatively affects prognosis, or
9. Results in temporary or permanent harm, disability, or death.

As per this definition, an allergic reaction (an immunologic hypersensitivity occurring as the result of unusual sensitivity to a drug) and an idiosyncratic reaction (an abnormal susceptibility to a drug that is peculiar to the individual) are also considered ADRs.[3]

Side effect: Unintended effect occurring at a normal dose related to the pharmacological properties.

Signal: Defined as reported information regarding a possible causal relationship between a drug and adverse events. Signal detection is an essential element of pharmacovigilance because it helps to identify previously unknown or incompletely documented ADRs. Based on the signal detection, regulators shall decide about possible regulatory actions.[4,5] The strength of the signal depends on the quality of information and the seriousness of the event. In any case, signal detection requires more than one report. To detect signals, pharmacovigilance relies on various sources and methods to establish a causal relationship between drug and an adverse event.

Serious ADE or ADR: Any untoward medical occurrence that at any dose

- results in death
- is life-threatening
- requires inpatient hospitalization or prolongation of existing hospitalization
- results in significant disability or incapacity.

Frequency of ADRs (CIOMS)[1]:

- Very common $\geq 1/10$
- Common (frequent) $\geq 1/100$ and $< 1/10$
- Uncommon (infrequent) $\geq 1/1000$ and $< 1/100$
- Rare $\geq 1/10{,}000$ and $< 1/1000$
- Very rare $< 1/10{,}000$

OVERVIEW OF PHARMACOVIGILANCE SYSTEMS

Pharmacovigilance systems are an integral part of healthcare policies in many jurisdictions. In the EU, both marketing authorization holders and member countries are responsible for monitoring the safety of authorized drugs.

Hence, pharmacovigilance in the EU is a system of shared responsibility and cooperation. Furthermore, effective pharmacovigilance systems require cooperation among various actors, such as regulators, pharmaceutical companies marketing a drug, healthcare professionals, and patients.

The cyclooxygenase 2 selective nonsteroidal antiinflammatory drug rofecoxib resulted in many early cardiovascular deaths in the 21st century, resulting in its withdrawal from the market. Improvements in numbers of quality reports and active participation of different stakeholders are essential for consumer safety. Moreover, global data and signal detection are useful in clinical practice throughout the world for patient safety. Advancement in communication in a current pharmacovigilance system shall decrease morbidity and mortality due to ADEs.

To improve the effectiveness of pharmacovigilance systems, ADRs should not be seen as personal failures or lead to the loss of reputation of healthcare professionals. Instead, a culture of care and patient safety should be embedded in national healthcare policies and medical practices in healthcare institutions. As part of this culture, ADR reporting is essential for pharmacovigilance. Gaps between knowledge and action should be decreased, given that response lag by manufacturers, regulators, and healthcare professionals on safety issues potentially result in more casualties.[6] Pharmaceutical products pose significant safety concerns in clinical trials and in clinical practice. Pharmacovigilance is one of the operations in any phase of the clinical trial and even in animal studies. Some safety research is not ethical (e.g., carcinogenicity studies), and thus is not carried out on humans. Once the medicinal product is approved and authorized to the market, the widespread use in bigger populations will help to build more safety data. No pharmaceutical products are without safety concerns; even natural foods can cause unintended effects in populations prone to risk. Healthcare professionals should identify these risk factors and prevent or minimize an actual untoward effect. Having an efficient system of pharmacovigilance in place is essential to consistently monitor and respond to safety concerns about pharmaceutical products.[7] However, some studies have found that underreporting of ADRs is widespread.[8] The main reasons for underreporting are lack of awareness of healthcare professionals regarding the importance of pharmacovigilance, the complexity of reporting, lack of cooperation among healthcare professionals, and technical hindrances.

Pharmacovigilance reports originally focused on reporting ADRs. Currently, with well-established pharmacovigilance systems, especially in industrialized nations, the reports include any drug-related problems. As per the "world medicines situation 2011 pharmacovigilance and safety of medicines," a pharmacovigilance center report includes the following:

- unexpected lack of efficacy;
- quality defects;
- drug abuse;
- medication errors;
- interactions with traditional and herbal drugs; and
- poisoning events.[9]

An ADE could happen due to the poor quality of the product or due to its improper use. It could also happen due to factors that are not under control. The extended focus on additional information is reflected in the EU pharmacovigilance definition, which includes medication errors, for instance.

Mainly, three factors could contribute to ADEs: poor product quality, suboptimal use, and unidentified factors (Fig. 15.1).

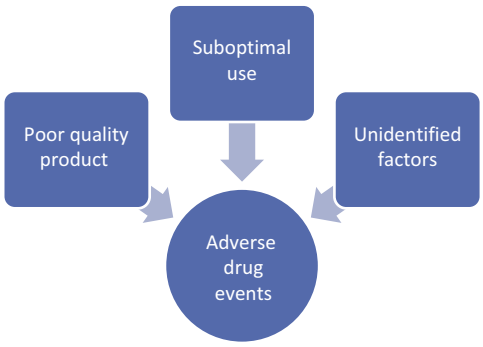


FIGURE 15.1 Possible factors resulting in adverse drug events.

Poor Quality Product

Reductions in quality may be intentional or unintentional. The quality of a product can be affected by various mishaps throughout the production cycle, including manufacturing, transportation, and storage. Poor quality assurance in terms of environmental control (temperature, humidity, air pollution, microbial contamination, etc.) can result in poor product quality. This is the case particularly with biological drugs (so-called biologicals). Biologicals are based on living cells and, therefore, are sensitive to change throughout the long and complex process of manufacturing. Adherence to quality standards is vital for ensuring consistent product quality and safety.

Finally, when the product is consumed by the patient, it may have reduced quantities of the labeled products or harmful substances. A good quality product should reach the consumer maintaining its quality and, when given to the patient, should produce intended effects and avoid preventable ADEs. In the United States, product problems should be reported to the FDA when there is a concern about the quality, authenticity, performance, or safety of any medication or device. In the EU, problems can be reported to various regulators in the member countries or directly to the European Medicines Agency (EMA). Problems with product quality include the following:

- suspect counterfeit product;
- product contamination;
- defective components;
- poor packaging or product mix-up;
- questionable stability;
- device malfunctions; and
- labeling concerns.

With drugs, a pharmacist is often the first to recognize a product quality problem. Nurses are often the first to recognize a problem with a medical device. A product problem should be reported through the pharmacovigilance system (e.g., MedWatch).

Suboptimal Drug Use

The scope of a pharmacist's service lies in the optimal use of pharmaceutical products. Rational use of drugs achieves better results. All aspects of drug use should be appropriate for the situation and the patient: right drug, right patient, right dose, right dosage form, right duration, right frequency, right combinations, right lifestyle, diet, etc. Pharmacists, in collaboration with policy-makers and other healthcare professionals, should establish the rational use of drugs.

Medication errors. Medication errors result in ADEs. A higher dose has higher chances of developing an ADE. A lower dose decreases the clinical effectiveness. As a result, the burden of disease is not reduced.

Drug interactions. Drug interactions are not necessarily clinically significant, and sometimes result in beneficial outcomes. However, drug interactions are often harmful and the potential for an ADR increases in the presence of another drug.

Unidentified Factors

ADEs could also happen due to idiosyncratic factors that cannot be explained by pharmacological logic. These ADEs, thus, are difficult to prevent. Confounding factors increase in the presence of comorbidities and multiple medications.

Characteristics of good pharmacovigilance systems include the following:

- Easy to use (reporting forms offline and also online communication like the Yellow Card App in the United Kingdom, accessible manuals for reporting forms, procedures for submission and collection of reports).
- Allows reporting by patients and healthcare providers. Even if healthcare providers and patients are legally allowed to report, easy to use forms and accessible manuals are essential for patients to report.
- Well-structured reports to facilitate analysis.
- Standardized procedures and definitions (e.g., what is a reportable event, follow-up, and processing of case reports).
- Allows analysis of product class level (e.g., erythropoietin) and on an individual product level. Regarding biologicals, reporting of batch numbers is essential given that biologicals are sensitive to change; previously unobserved ADEs may occur only for specific batches of the same drug. However, studies have shown that batch numbers are often not reported even in industrialized nations.
- Procedures for analysis of aggregated information.

- Efficient communication practices improve awareness to report ADRs. Underreporting of ADRs by healthcare professionals and particularly patients is common. All possible means of communication should be utilized, e.g., websites, social media, leaflets, etc. In some countries, regulators cooperate with schools to strengthen the national pharmacovigilance system in the long run.
- Training of healthcare professionals. Training could improve reporting of ADRs (i.e., quantity) and the quality of reported information. E-learning tools and educational materials for healthcare professionals and patients are useful to increase the quantity and quality of ADR reporting.

OBJECTIVE 15.2. DESCRIBE THE PHARMACOVIGILANCE FRAMEWORK AND THE ROLES OF DIFFERENT STAKEHOLDERS IN PATIENT SAFETY

Pharmacovigilance systems are based on the shared responsibilities of different stakeholders, including international organizations, healthcare institutions, policy-makers, and patients. With a view to the effective operation of these systems, different stakeholders should be connected for mutual benefits, and roles and responsibilities should be well defined. In the EU, for instance, the European Network of Centres for Pharmacoepidemiology and Pharmacovigilance aims to improve the science and practice of pharmacovigilance by connecting various stakeholders. Rare ADRs happen sporadically. The rest of the world benefits from reporting of such information. In resource-limited countries, integrating supporting programs are essential for building capacity in pharmacovigilance systems.[20]

Fully developed pharmacovigilance systems and responsible actions of all stakeholders could prevent and manage more ADEs than otherwise. Effective and timely communication between different stakeholders is a key aspect to share and utilize scientific information pertaining to pharmacovigilance. Other specific actions by different stakeholders in pharmacovigilance are provided in the following sections.[11]

Patient/Carers

Patients or their carers report ADEs to healthcare professionals. This includes mostly pharmacists, doctors, and nurses. Patients have limited skills to differentiate between adverse events due to drugs or other factors. Consulting healthcare professionals helps in clarifying what is going wrong.

Healthcare Professionals

Many times, healthcare professionals may not report such events to pharmacovigilance systems due to various constraints, including busy schedules. Depending on the national jurisdiction, in which the pharmocovigilance system operates, they might also fear official inquiries and legal consequences, in case they are held liable for ADEs. The WHO in its guidelines on patient safety states that "the fundamental role of patient safety reporting systems is to enhance patient safety by learning from failures of the healthcare system. Reporting must be safe. Individuals who report incidents must not be punished or suffer ill-effects from reporting." Effective policies on patient safety require sound legal foundations, but there are also social and cultural challenges in healthcare institutions in the daily practice of pharmacovigilance.[21]

National pharmacovigilance systems have different terms and definitions with regard to the notion of healthcare professionals, but the following groups can be included: medical doctors (including general and specialized practitioners), nursing and midwifery professionals, dentists, and pharmacists.

The widespread underreporting of ADRs, among other reasons, is due to ADRs being often seen as a personal failure and associated with a loss of reputation. Apart from social changes in healthcare institutions, effective pharmacovigilance requires a no-blame culture in which healthcare professionals can report ADEs without having to fear legal consequences. The Council of Europe advocates the development of pharmacovigilance systems in which it is not the individual that is blamed but that aims to improve the institutional conditions under which healthcare professionals work.[22] Hence, in addition to the legal foundation of such systems, effective pharmacovigilance also faces social and cultural challenges regarding the daily practice in healthcare institutions.[21]

Healthcare professionals are supposed to be trained to take part efficiently in pharmacovigilance operations. The WHO states that healthcare professionals "maintain health in humans through the application of the principles and procedures of evidence-based practice and caring."[23]

They also have responsibilities to update their knowledge in the field through continuing education. Getting to know newer safety information about pharmaceutical products greatly benefits their patients and increases patient safety. Technical knowledge and skills in assessing patient conditions help healthcare professionals to monitor patient outcomes

after medication use. Patients are motivated to report their problems to the healthcare professionals to get clarity on what is going wrong. Nurses and pharmacists usually report any adverse events, which they are not clear about, to the doctors who have higher skills in diagnosing an ADR. Well-educated pharmacists with clinical skills are in a better position than ever before in diagnosing many ADEs by themselves.

Manufacturers

From the viewpoint of trade, as drugs are manufactured and sold by the pharmaceutical companies, they are ultimately responsible for quality and safety of the product. In the pharmacovigilance policy of the EU, for instance, companies holding authorizations to market drugs are responsible for operating a pharmacovigilance system. This system operates in concert with the pharmacovigilance system of the member countries. However, the responsibility of the pharmaceutical companies as the authorization holder does not mean that other stakeholders are not responsible. Manufacturers have to work with healthcare professionals, patients, and regulators to generate safety reports about pharmaceutical products.

Regulators

Drug regulators make laws, guidelines, and policies related to manufacturing, distribution, storage, use, and disposal of pharmaceutical products at all levels of government. At the international level, pharmacovigilance is coordinated by the WHO. At the regional level, the EU has developed a comprehensive pharmacovigilance policy in close cooperation with national regulators. Regulators approve pharmaceutical products to market when the reports suggest its acceptable safety profiles. This can be referred to as market authorization. The regulators continue to monitor the product and mandate the manufacturers to report ADEs from clinical practice. Hence, marketing authorization holders are key actors of pharmacovigilance. Regulators also welcome voluntary reports from healthcare professionals and patients on safety issue experiences related to the products.

Regulators usually establish a pharmacovigilance system with a framework of national, regional, and institutional pharmacovigilance centers. Across levels of government, coordination and networking are crucial for effective pharmacovigilance. Technical and financial support is provided to these centers for generating and communicating quality reports. In the EU, the EMA is at the hub of national regulator's networks. The main task of EMA is to coordinate the evaluation of medicinal products and to advise the EU institutions and the member countries on issues relating to regulation. Following the latest reform of drug policy in the EU, the Pharmacovigilance Risk Assessment Committee (PRAC) has been created in the EMA framework. The PRAC assesses and monitors the safety of medicinal products in Europe.

In most pharmacovigilance systems, like in the United States, regulators also monitor international plants that manufacture pharmaceutical products to be used in the country. Regulators create and maintain pharmacovigilance systems in their countries and coordinate activities of different stakeholders within that country. The national pharmacovigilance systems connect to international pharmacovigilance systems. Pharmacovigilance activities happen through regular communication to manufacturers, healthcare professionals, and patients in the country.

Regulators are committed to train healthcare professionals and encourage clinician and patient reporting of ADEs. Governments/regulators should have sufficient technical and financial recourses to develop and maintain effective pharmacovigilance systems. However, even in industrialized nations, the effectiveness of pharmacovigilance is hampered by the lack of resources and training.[8]

International Organizations and Healthcare Organizations

Organizations like WHO coordinate reporting of drug-related ADEs and generate global information that is shared with different stakeholders. WHO also takes a major role in ensuring the quality of drugs manufactured and used. It has measures to standardize drugs globally by assigning generic names and promoting generic prescribing. Assigning anatomical, therapeutic, and chemical coding and defined daily dose is another strategy to identify and standardize the use of drugs by WHO.

In the EU, the EMA coordinates the evaluation of drugs by national regulators. EMA also maintains the collection of ADR reports in the EudraVigilance database. Through various networks and groups of experts and regulators, EMA contributes to the development of EU health policy in general and pharmacovigilance in particular.

Organizations like ISMP, universities, and healthcare facilities conduct research to generate pharmacovigilance information. The information is communicated to other stakeholders directly or through publishing in appropriate media.

Most of the organizations start their pharmacovigilance centers to perform their operations. Drug and poison information centers in healthcare facilities sometimes coordinate pharmacovigilance operations as an additional service.

These organizations also organize training programs for healthcare professionals or patients/caregivers in different aspects for safe use of pharmaceutical products. Organizations also develop guidelines for the healthcare professionals to implement and monitor the safe use of drugs.

Pharmacists should take care to maintain the quality of a product, prescribers should recommend it appropriately, the nurse should administer it safely, and the patient/caregivers should follow all instructions provided by healthcare professionals. Pharmacists, being specialists of drugs and with clinical expertise, should monitor all stages of drug use for the benefit of the patients.

The notion of patient safety has gained traction in the last few decades to become an issue of high salience in public health and healthcare. Various individual and institutional factors drove the development to patient safety: committed healthcare professionals and developments in public policy making, risk management, and legal systems. Accordingly, policies on patient safety entail multiple challenges for legal systems and the management and culture of healthcare institutions.[21] Pharmacovigilance and reporting of ADEs is a key component of public health and healthcare.

Pharmacovigilance is an essential part of the healthcare system, and ensuring public health through product safety is of great importance in industrialized countries. However, there is significant variation in terms of organization, financing, and responsibility for health. Accordingly, health policy and pharmacovigilance systems vary even in rather homogeneous regions such as Europe, for instance.[8] Given that pharmacovigilance relies on cooperation and coordination at the international level, international organizations are important for developing patient safety policies.

At the international level, the WHO is a key actor in promoting and implementing patient safety policies. In 2002, the 55th World Health Assembly adopted a resolution urging Member States to "pay the closest possible attention to the problem of patient safety."[24] The resolution emphasized the importance of global norms and standards for product safety, ADR reporting, and a culture of safety in healthcare organizations. In the wake of the resolution, two Patient Safety Global Action Summits were held in 2016 and 2017 to facilitate exchange and collaboration between medical experts and policymakers. Such collaboration is needed to address the multiple challenges of pharmacovigilance.

A key issue is the development of legal systems that facilitate the reporting of ADEs. The WHO and the Council of Europe should work toward promoting the development of legal systems facilitating ADR reporting. For many healthcare professionals, reporting of ADRs is seen as a personal failure and associated with a loss of reputation. Apart from the social and cultural challenges in healthcare institutions, effective pharmacovigilance requires a legal system that facilitates reporting. The WHO guidelines for ADE reporting state that patient safety builds on patient reporting and that individuals who report must not face negative consequences.[25] The Council of Europe has issued recommendations to balance the requirements of patient safety through reporting and learning, and the legal protection of patients with confidentiality and legal protection of healthcare professionals.[22] This is referred to as a culture of no blame. Some countries have introduced legislation to facilitate patient reporting in a no-blame culture.[21]

To improve patient safety in nonindustrialized countries, the WHO, together with public and private donors, promotes and implements additional methods of pharmacovigilance to complement spontaneous ADR reporting.[26]

Joint Commission International accreditation of healthcare facilities includes monitoring on the International Patient Safety Goals (IPSGs). IPSGs help accredited organizations address specific areas of concern in some of the most problematic areas of patient safety. The six IPSGs are[27]

Goal 1: Identify patients correctly
Goal 2: Improve effective communication
Goal 3: Improve the safety of high-alert medications
Goal 4: Ensure safe surgery
Goal 5: Reduce the risk of healthcare-associated infections
Goal 6: Reduce the risk of patient harm resulting from falls

OBJECTIVE 15.3. EXPLAIN HOW TO REPORT ADVERSE DRUG EVENTS

ADRs could be clinical or variations in laboratory values. Different trigger tools help in reporting. Standardized forms are used in reporting ADEs. These forms are designed for easy filling and for compliance with the format used in the international pharmacovigilance system. The national and institutional ADE reporting format is standardized in international format. After the ADE data are collected, they are analyzed for causality, severity, preventability, etc. The WHO's International Drug Monitoring Program collects Individual Case Safety Reports (ICSRs) from its member states. Healthcare professionals could report ADEs to regional or national pharmacovigilance centers. If a causal relationship

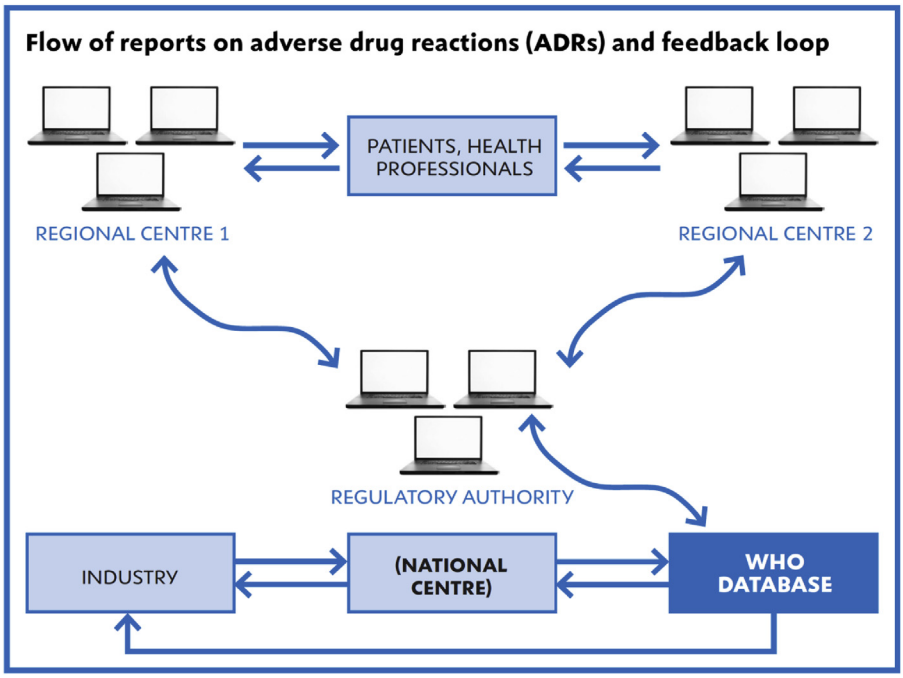


FIGURE 15.2 Flow of reports on adverse drug reaction and their feedback. *Adapted with permission from Pal S, Dodoo A, Mantel A, Olsson S.* The World Medicines Situation 2011. Pharmacovigilance and Safety of Medicines. *Geneva: World Health Oraganization; 2011. http://apps.who.int/medicinedocs/documents/s18771en/s18771en.pdf.*

is established, it will be considered as an ADR, if not an ADE. Many regulators consider the reporting system to be for ADEs as it is advisable to report any adverse events even if the causal relationship with a drug is not established. Consumers might report adverse experiences after consuming their medication. They are not trained to conclude the causal relationship. All that they are submitting are ADEs until the causal relationship is established by the experts. National pharmacovigilance centers are usually part of the country's drug regulatory agency. The national center will forward these reports to the WHO collaborating center for International Drug Monitoring, Uppsala Monitoring Center (UMC) in Sweden. The flow of reports on ADRs and their feedback is illustrated in Fig. 15.2.

Vigiflow. Vigiflow is software from the UMC for the reporting of ADRs. It helps, especially, developing countries with limited technical resources to report ADRs.

Vigiaccess. Vigiaccess is an open access database from the UMC. Anyone can access it to see the numbers of ADR reports globally.[28]

Spontaneous Reporting on Adverse Drug Reactions

Most countries, industrialized and nonindustrialized alike, rely primarily on spontaneous and voluntary ADR reporting.[26] Spontaneous reporting of suspected ADRs (these are ADEs suspected as ADRs; confirmed as ADRs when causality relationship is established) is a voluntary activity involving any healthcare professionals or consumers. Many safety signals are generated through spontaneous reporting, encouraging voluntary reporting results in a safety culture. All stakeholders communicate through a pharmacovigilance system that reminds everyone of their responsibilities in detecting, preventing, and managing ADEs. With voluntary reporting, the numbers of reports generated are relatively few compared with the actual occurrence of ADEs. The reports could also be incomplete. Most of the reports are based on suspicions based on ADEs that happen following drug use. It is not clear that those cases are truly an ADR to the reported drug. Many factors affect voluntary reporting of suspected ADRs:

- Diffusion of responsibility (many professionals feel someone else shall report the event, and finally no one is responsible for taking it forward).
- Technical insufficiencies for reporting to the pharmacovigilance system.
- Fear of punishment or blame for harming the patients.
- Time constraints in reporting and follow-up if any.

- Institutional policies that discourage reporting of ADEs due to concerns for reputation.
- Lack of training and motivation to healthcare professionals shall decrease abilities to detect ADRs or ignore them.[11]

Examples of databases that collect voluntary reporting:

- WHO International Drug Monitoring Program Vigibase
- US FDA ADR Reporting System for Pharmaceutical Products (FAERS)

Some countries use registries instead of reporting to such databases. These registries from databases could be used further in epidemiological studies. Many hospitals keep their own registries, like a cancer registry, pregnancy registry, ADE registry, etc. Registries are helpful for the organization to analyze the trends in ADEs and become a quality tool to observe the effectiveness of remedial actions to decrease ADEs. Registries are not a good source of rare events as there will be only a few or none. Instead, if these registries connect with international databases that pool events reported from a bigger population, they allow the gathering of more generalizable information. The problem of connecting registries to the international database is its compliance with the formats.

Mandatory Reporting

It is compulsory for manufacturers to conduct postmarketing surveillance of their products in most countries. The pharmacovigilance report should be submitted to regulators at prescribed intervals. When a new drug is approved for marketing, for a few years it is mandatory to report ADEs to the regulators. Manufacturers provide technical and financial support to healthcare facilities to enable reporting. They also facilitate training to healthcare professionals for reporting. Certain national healthcare systems integrate pharmacovigilance reporting into clinical practice. The presence of these types of systems in developed countries results in higher reporting. Such mandatory reporting is present in some of the public health programs (HIV/AIDS, malaria, tuberculosis, etc.) in developing countries. External funding, technical support, and frequent training improve reporting.[11]

EudraVigilance is not exactly voluntary. Marketing authorization holders must report ADRs during drug development after marketing. Reporting obligations are defined in the respective EU legislation and guidance.

Features of a good pharmacovigilance report[29]:

- Description of ADEs
- Suspected and concomitant product therapy details (e.g., dose, dates of therapy)
- Patient characteristics (e.g., age, sex), baseline medical condition, comorbid condition, family history, other risk factors
- Documentation of the diagnosis
- Clinical course and outcomes
- Relevant therapeutic measures and laboratory data
- Dechallenge and rechallenge information
- Reporter contact information
- Any other relevant information

CONCLUSION

Pharmacovigilance is a part of healthcare systems worldwide. The WHO leads pharmacovigilance operations and provides technical support in reporting ADRs. Many countries have well-built pharmacovigilance systems, but actual incidence of ADRs is much higher than what is reported. Underreporting of ADRs is a major problem as well as the quality of reports. The basic objective of pharmacovigilance is the safe use of drugs, patient safety, and, ultimately, safeguarding public health. To achieve this goal, national regulators and international organizations should empower healthcare professionals and the public to report more ADRs.